

Scaling Neuroevolution for Heterogenous Tasks using Universal Knowledge Representation

Neuroevolution to Simultaneously Execute Heterogeneous Tasks

Romain Claret at DEBS'23 (Ongoing Research)

1 Intro

- Nature-inspired NEAT variation
- Universal Knowledge Format
- Perform Heterogeneous Tasks

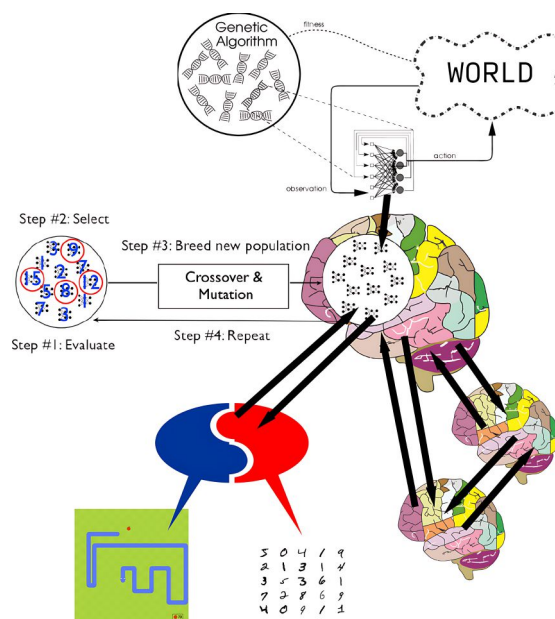
2 Objectives

- Extend NEAT for context-free heterogeneous tasks
- Combine Artificial Intelligence and Neuroscience research
- Explore scalable Artificial Collective Intelligence for complex environments

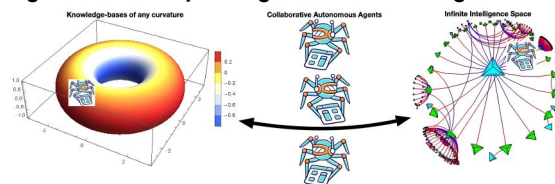
3 Methodology

- Evaluate on task-specific, and heterogeneous tasks
- Compare with x-NEAT and Deep-learning models

4 Current Architecture

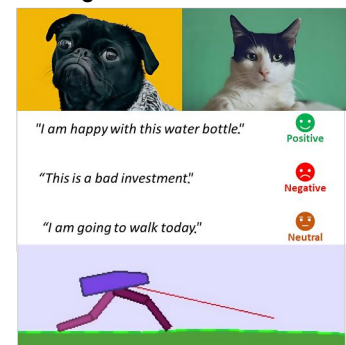


Agents collectively waving Universal Knowledge

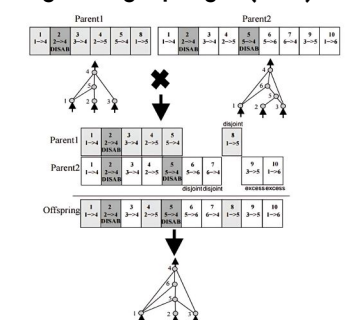


Extra info

Heterogeneous Tasks



Evolving Neural Networks through Augmenting Topologies (NEAT)



Research Question:

What is an optimal approach for a universal knowledge representation in neuroevolution for various heterogeneous tasks with fixed input and output neurons?

GEENNS Framework (Graph-Embedded Evolving Neural Networks Synergy)

Universal Knowledge Representation, Multitask Learning, Prediction Consensus, Continual Learning, and more...



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